Hyperspectral Image Denoising by Asymmetric Noise Modeling

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***Abstract*— In general, hyperspectral images (HSIs) are degraded by a mixture of complicated noise (i.e., mixture of Gaussian and sparse noise), and how to precisely model HSI noise plays a vital role in the task of HSI denoising. The most popular choices for encoding the noise distribution are Gaussian, Laplacian, and the mixture of Gaussians, but they are always incompatible with real-world HSI noise. By investigating histograms of the error map, we first explore that asymmetry is a typical and general feature of HSI noise. Inspired by this discovery, we find that a bandwise asymmetric Laplacian (AL) distribution can be finely used to model this type of noise. Equipped with the low-rank matrix factorization (LRMF) framework, we formulate a novel model by the maximum likelihood estimation (MLE) principle, which can be efficiently solved using the iterative optimization algorithm. Extensive experimental results on synthetic and real datasets demonstrate that the proposed model outperforms other counterparts. It is also found that scale and asymmetry parameters in the AL distribution can well interpret the pattern of real-world HSI noise.**

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***Index Terms*— Asymmetric Laplacian (AL) distribution, hyperspectral image (HSI) denoising, low-rank matrix factorization (LRMF).**

I. Introduction

DUE to the high spectral resolution, hyperspectral images (HSIs) can provide relatively comprehensive characterizations for the Earth's surface [1], and thus it is widely applied in the field of remote sensing, e.g., classification [2] and objection detection [3]. Nonetheless, due to the limitation of imaging devices, the obtained HSIs always contain complex noise which absolutely hinders HSIs' subsequent applications. Therefore, HSI denoising has attracted much attention in recent years.

The natural way for HSI denoising task is to apply the traditional 1-D or 2-D denoising methods band by band (e.g., sparse dictionary representation [4] and nonlocal sparse representation [5]) or generalize the 1-D or 2-D denoising methods as the 3-D one (e.g., 3-D Wiener filters [6]). Since these methods largely ignore the spectral correlation of HSI data, the results are often unsatisfactory.

Recently, low-rank matrix factorization (LRMF) has been developed as a powerful technique for HSI denoising. It is well-known that many singular values of a clean HSI approach zeros, so this low-rank property is expected to deliver useful prior knowledge for denoising. Given an HSI with B-bands and $M \times N$ pixels denoted by $\boldsymbol{Y} \in \mathbb{R}^{MN \times B}$, LRMF approximates it by the product of two factor matrices

$$\min_{\boldsymbol{U} \in \mathbb{R}^{MN \times R}, \boldsymbol{V} \in \mathbb{R}^{B \times R}} \left\| \boldsymbol{Y} - \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} \right\|_2^2. \tag{1}$$

The solution is given by the truncated singular value decomposition (SVD) [7]. However, the ℓ_2-norm is only optimal for the Gaussian noise, and it is always sensitive to complex noise essentially contained in real HSIs [8]. To cope with the problem, robust principal component analysis (RPCA) assumes that a noisy HSI $\boldsymbol{Y}$ can be decomposed into the low-rank component $\boldsymbol{L}$ plus sparse noise $\boldsymbol{S}$, corresponding to the following relaxed optimization issue:

$$\min_{\boldsymbol{L}, \boldsymbol{S}} \|\boldsymbol{L}\|_* + \lambda \|\boldsymbol{S}\|_1, \quad \text{s.t. } \boldsymbol{Y} = \boldsymbol{L} + \boldsymbol{S} \tag{2}$$

where $\|\boldsymbol{L}\|_* = \sum_i \sigma_i(\boldsymbol{L})$ denotes the nuclear norm, defined by the sum of singular values $\sigma_i(\boldsymbol{L})$. The above equation uses the ℓ_1-norm as the loss function and is more robust to severe noise than the ℓ_2-norm. Literature [9], one of the early work applying RPCA to HSI denoising, has attained impressive performance and validated this point. It is worth noting that the underlying assumption of the ℓ_1-norm loss function is that noise obeys the Laplacian distribution. Thus, RPCA is also not

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Fig. 1. Analysis on real HSI noise. Band 76 of the Urban dataset is selected for illustration. (d) Probability density function curve fit by AL distribution is displayed, and a gray box covers the interval $[-3\sigma, 3\sigma]$ (here, σ is the standard deviation of noise). Evident asymmetry and heavy-tailness can be observed in the circles where local areas of the noise histogram are amplified. (a) Noisy band. (b) Clean band. (c) Noise map. (d) Noise histogram.

universally optimal and may lead to unpleasant results when faced with non-Laplacian noise.

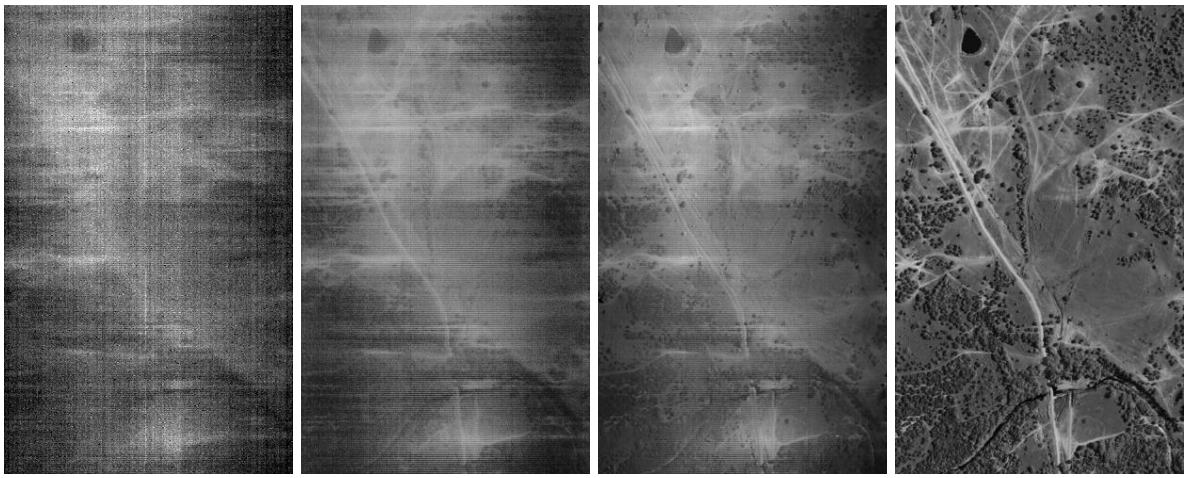
This dilemma urges researchers to explore noise modeling for HSI denoising. To alleviate such assumption incompatibility on real-world noise, Meng and De La Torre [8] propose to model the noise as a mixture of Gaussians (MoG) which is the theoretically universal approximation to any continuous distributions. In other words, MoG modeling is able to handle complex HSI noise with an unknown statistical distribution. Some variants of MoG modeling further verify its application on HSI denoising [10], [11], [12]. However, all these models assume noise identically and independently distributed (i.i.d.), which largely underestimates the insightful non-i.i.d. properties underlying real HSIs. Chen et al. [13] first observe that the noise distributions of different band images are statistically disparate but have a certain correlation, and then propose the non-i.i.d. MoG (NMoG) modeling to describe this phenomenon. These works have made remarkable progress and deepened understanding on HSI denoising. However, the success of MoG and its variants essentially lies in the universal approximation property, which enables MoG to provide a black-box way to model unknown noise. Are there other strategies to precisely model complicated HSI noise? It is an interesting but challenging problem.

Against the issue, this study attempts to provide more insights on HSI noise modeling by empirically exploring the noise histogram. First, we synthesize noisy HSIs corrupted by different kinds of mixed noise, including Gaussian, impulse, deadlines, and stripes' noise. It is found that the synthetic mixed noise is heavy-tailed and asymmetric, as analyzed in Fig. S1 of *Supplementary Material*. This observation motivates us to explore whether HSI noise in the real-world has the same pattern. HSI noise is inaccessible in practice, since a sensor cannot provide a pair of noisy and clean HSI. Due to HSI's spectrum continuity, a clean band can be approximately obtained by averaging neighbor bands around the center band, when these bands are corrupted by mild noise. This strategy is applied to band 76 of the Urban dataset. Five bands (i.e., bands 74, 75, 76, 77, and 78) are selected to generate an approximately clean band. Finally, the noisy band minus the approximately clean band is regarded as the noise map. As shown in Fig. 1(a), this noisy band contains a vertical deadline and a number of horizon stripes, and it is found that the extracted noise map in Fig. 1(c) matches our observation from Fig. 1(a). Therefore, the extracted noise map is in line with our intuition.

Let us focus on the noise histogram in Fig. 1(d) and two observations can be attained: first, it is observed that the noise frequency $f(n)$ is high when noise n is beyond the interval $[-3\sigma, 3\sigma]$ (σ is the standard deviation of noise) marked by a gray box. Hence, the noise is heavy-tailed. This observation has been widely validated in numerous previous works [9], [14], [15]. Second, it is found that the noise is highly asymmetric. For instance, the areas around $n = -0.08$ and 0.08 are amplified in the histogram. If this noise distribution is symmetric, the noise frequencies $f(n)$ and $f(-n)$ should be equal for any $n \in \mathbb{R}$, but evidently $f(-0.08)$ and $f(0.08)$ have a great gap. This observation should be potentially useful for more precisely establishing noise models, but still have not been presented in HSI denoising. In previous works, the Laplacian and MoG are only able to model the heavy-tailness. Due to the fact that HSI noise is both heavy-tailed and asymmetric, we generalize the Laplacian distributions as asymmetric Laplacian (AL) distributions to model both heavy-tailness and asymmetry. Probability density functions marked by the solid lines in Fig. 1(d) show that AL distributions can well fit HSI noise. Besides this image, more examples shown in Figs. S2–S6 of *Supplementary Material* also support the above observation. In conclusion, HSIs contain both symmetric noise and a large amount of asymmetric noise, and thus modeling the unique HSI noise distribution is a key factor to improve the performance of HSI denoising.

Considering that HSIs suffer from both the symmetric and asymmetric noise distributions, *we assume that the noise of each band obeys an asymmetric noise distribution* and propose a novel noise modeling framework to describe the actual noise distribution of HSIs. Remark that the (symmetric) Laplacian noise distribution can be expressed as a special case under the AL noise distribution, so the proposed method is a universal framework that models not only symmetric but also asymmetric noise. Based on this framework, a bandwise AL noise-based matrix factorization (BALMF) model is proposed. The parameters of this model can be estimated by the principle of maximum likelihood estimation (MLE). To validate BALMF's performance, SVD, low-rank matrix recovery (LRMR) [9], and BALMF are applied to a real HSI dataset, i.e., Terrain, and the denoised images are shown in Fig. 2. SVD and LRMR assume that noise comes from the Gaussian

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(a) (b) (c) (d)

Fig. 2. Denoised results obtained by SVD, LRMR, and BALMF on the Terrain dataset. (a) Noisy band. (b) SVD. (c) LRMR. (d) BALMF.

and Laplacian distributions, respectively. BALMF evidently obtains a superiority on removing noise and recovering a clean image, demonstrating its better noise modeling capability than the conventional Gaussian- or Laplacian-based noise modeling methods. Furthermore, a series of experiments conducted on two synthetic and two real datasets demonstrate that BALMF achieves better and comparable results, compared with ten state-of-the-art (SOTA) methods.

The rest of this article is organized as follows. Section II reviews the related work. Section III introduces the proposed model, and a series of experiment are conducted in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Low-Rank Models

On the basis of standard low-rank models (1) and (2), the following works mainly focus on developing more rational regularization (a.k.a. prior knowledge) or data fidelity (a.k.a. noise modeling).

1) Prior Knowledge for HSI Denoising: Three kinds of prior knowledge are frequently used in HSI denoising, including low-rankness, local smoothness, and nonlocal similarity (NLS).

Low-rankness refers to the fact that HSIs should lie in a low-dimensional spectral subspace, since the images in adjacent bands are usually collected with similar sensor parameter settings. It can be encoded by low-rank matrix/tensor factorization or nuclear norm [16]. For LRMF, the low-rank component is represented by $\boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}$; for low-rank tensor factorization, the low-rank component can be represented by different decomposition formats, such as CANDECOMP/PARAFAC and Tucker decomposition [7], [17], [18]. For RPCA and its variants, the nuclear norm $\|\boldsymbol{L}\|_*$, known as the convex relaxation of matrix rank, is used to model the low-rankness of HSIs. It is defined by the sum of singular values of a matrix, $\|\boldsymbol{L}\|_* = \sum_i \sigma_i(\boldsymbol{L})$, where $\sigma_i(\boldsymbol{L})$ denotes the singular value of matrix $\boldsymbol{L}$. Chen et al. claim that larger singular values contain more information and should be less penalized [19]. They hence use the nonconvex γ-norm $\|\boldsymbol{L}\|_{\gamma,*} = \sum_i (1 - e^{-\sigma_i(\boldsymbol{L})/\gamma})$ to approximate rank. Zeng et al. [20] generalize γ-norm to tensor γ-norm, and apply it to the tensor RPCA model. In the following works, it is reported that other nonconvex norms also lead to good performance on the task, such as Schatten p-norm [21], [22], log-determinant rank approximation [23], and normalized ϵ-penalty [24].

The local smoothness refers to the fact that pixels in a local area tend to have similar values. This is the commonly used prior for images, which is often characterized by the total variation (TV) regularization and has been integrated into the low-rank approximation framework [14], [15], [25], [26], [27]. LRTV developed by He et al. [14] is a typical model along this research line. They define a hyperspectral TV (HTV) as the sum of TV over all the bands. Some variants of HTV have been explored to prevent oversmoothness and artifacts. With the similar idea of HTV, Yang et al. [25] extend Moreau-enhanced TV for HSI. Cai et al. [28] present a weighted graph TV to model the smoothness between neighboring pixels. Intending to preserve more details, Wang et al. [29] replace the ℓ_1-norm with the ℓ_0-norm to penalize image gradient magnitudes. Furthermore, Wang et al. [30] suggest to combine the ℓ_1- and ℓ_0-norms to overcome their shortcomings. Peng et al. [31] further propose an enhanced 3-D-TV regularization to model the sparsity on gradient maps of an HSI.

NLS refers to another intrinsic HSI prior structure that similar structures and patterns repeat in the whole scene of images [32]. Given an HSI represented by $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{M\times N\times B}$, the NLS-based methods first reshape $\boldsymbol{\mathcal{Y}}$ as a group of 2-D fullband patches $\boldsymbol{\Omega}_{\boldsymbol{\mathcal{Y}}} = \{\boldsymbol{Y}_i \in \mathbb{R}^{mn\times B}\}_{i=1}^{P}$, where P is the number of patches, and m and n are the height and width of the patch, respectively. Then, for a patch $\boldsymbol{Y}_i$, block matching is applied to find its nonlocal similar patches, and they are stacked as a third-order tensor $\boldsymbol{\mathcal{Y}}_i \in \mathbb{R}^{mn\times B\times p}$, where p denotes the number of nonlocal similar patches. The stacked patch is also low-rank and can be denoised by different kinds of low-rank algorithms. Finally, a denoised HSI

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is reconstructed by aggregating all the denoised patches. Most NLS-based methods focus on how to denoise the stacked patch $\mathcal{Y}_i$, such as tensor dictionary learning [33], intrinsic tensor sparsity regularization [34], Laplacian scale mixture modeling [35], hyper-Laplacian spectral gradient prior [36], Tucker decomposition [37], and tensor ring decomposition [38]. However, the large spectral number tends to bring high computational burden. To alleviate this problem, He et al. [39] integrate the global spectral low-rank property and NLS and develop a fast NLS-based algorithm, where NLS denoising is applied to reduced images which are obtained by low-rank decomposition.

2) Noise Modeling for HSI Denoising: In addition to prior knowledge, noise modeling also plays an important role in HSI denoising. The ℓ_2-norm is a popular choice for data fidelity, and it intrinsically corresponds to Gaussian noise distribution assumptions. The ℓ_2-norm's sensitivity to outliers makes it hardly to be performable on real-world HSIs. By virtue of the robustness to outliers, the ℓ_1-norm loss term becomes prevalent and has achieved good performance especially in heavy noise scenarios [40]. However, the ℓ_1-norm of the noise term hypothesizes that noise obeys the Laplacian distribution, and real-world HSI noise is incompatible with the ℓ_1-norm's hypothesis.

Meng and La Torre [8] develop the MoG-based LRMF model (MoG-LRMF). The universal approximation property of MoG makes the model able to adaptively fit noise with unknown statistical distributions. This idea has also been applied to low-rank tensor factorization [41], [42], [43]. However, MoG could approximate a complex distribution (e.g., Laplacian distribution) only if the number of Gaussian components goes to infinity, and thus it is infeasible in practice. Against this issue, Cao et al. propose a mixture of exponential power distributions (MoEP) [11], [12] to model unknown noise and also design a penalized way to adaptively select the mixture component number. The component in MoEP is very complicated and it could alleviate MoG's deficiency to some extent. Chen et al. claim that HSI noise is non-i.i.d., because only a small part of bands is corrupted by strong noise, while others are with relatively weak noise or approximately noise-free. Due to this fact, they build a non-i.i.d. MoG (NMoG) framework [13], where each band is assigned with different forms of MoG noises. Nonetheless, all these noise modeling methods ignore the fact that HSI noise is asymmetric.

B. Deep-Learning-Based Methods

Recently, convolutional neural networks (CNNs) have been an influential tool for HSI denoising. Different from the low-rank models, CNNs learn a complicated nonlinear function to map noisy HSIs to clean ones on massive training datasets. Two seminal HSI denoising networks composed of 2-D filters are built in [44] and [45]. Dong et al. [46] point out that these 2-D CNNs cannot make full use of strong spectral correlations, and thus design a 3-D U-shaped network to learn spatial–spectral representations. In the following work, a number of researchers are devoted to enhance the representation capability of HSI denoising CNNs, e.g., improving receptive field by dilated convolution [47] and decreasing redundant information by attention [48], [49], [50], [51]. Although CNNs are powerful enough to fit training datasets, they are criticized for poor generalization capability, especially on complicated real non-i.i.d. noise cases.

III. Methodology

A. Model Formulation

This section proposes a BALMF model. Given an HSI with B bands and $M \times N$ pixels denoted by $\boldsymbol{Y} \in \mathbb{R}^{MN\times B}$, it can be represented by the product of two factor matrices ($\boldsymbol{U} \in \mathbb{R}^{MN\times R}$ and $\boldsymbol{V} \in \mathbb{R}^{B\times R}$) plus the noise $\boldsymbol{N} \in \mathbb{R}^{MN\times B}$

$$\boldsymbol{Y} = \boldsymbol{X} + \boldsymbol{N} = \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} + \boldsymbol{N}. \tag{3}$$

Equivalently

$$y_{ij} = x_{ij} + n_{ij} = \boldsymbol{u}_{i,:}\boldsymbol{v}_{j,:}^{\mathrm{T}} + n_{ij} \tag{4}$$

where $\boldsymbol{u}_{i,:}$ denotes the ith row of $\boldsymbol{U}$, and $\boldsymbol{v}_{j,:}$ denotes the jth row of $\boldsymbol{V}$.

As stated in Section I, HSI noise is heavy-tailed and asymmetric, and different bands are statistically disparate (i.e., not identical) but have a certain correlation (i.e., not independent). The proposed model hence assumes that each band is corrupted by AL noise

$$n_{ij} \sim \mathrm{AL}_j\left(n_{ij}\middle|0, \lambda_j, \kappa_j\right) \quad \forall j = 1, 2, \ldots, B. \tag{5}$$

The probability density function of the AL distribution is

$$p_{\mathrm{AL}}(x; \mu, \lambda, \kappa) = \lambda\kappa\,(1-\kappa)\exp\left(-|x-\mu|\,\lambda\eta\right) \tag{6}$$

where μ, λ, and κ are the location, scale, and asymmetry parameters, respectively. $\eta = \kappa\mathbb{I}(x \geqslant 0) + (1-\kappa)\mathbb{I}(x < 0)$, and $\mathbb{I}(x \geqslant 0)$ is an indicator function taking 1 if $x \geqslant 0$ and 0 otherwise. The skewness for AL distributions is $(2(\kappa^3 - (1-\kappa)^3))/(((1-\kappa)^2+\kappa^2)^{3/2})$ ranging from -2 to 2 [52].

For the scale parameter λ, it can be regarded as the inverse of variance. In other words, a smaller scale parameter λ results in a bigger variance. The asymmetry parameter κ ranges from zero to one. The AL distribution is positive skew when $0 < \kappa < 0.5$, and negative skew when $0.5 < \kappa < 1$. *By setting $\kappa = 0.5$, the AL distribution is then degenerated to the form of the Laplacian distribution. Thus, the AL distribution is also compatible with symmetric noise, although we always highlight that AL distribution is designed for asymmetric noise in the above passages.* The experimental results described in Section IV-C also show that the estimated κ approaches 0.5 when the noise is approximately symmetric and does not equal to 0.5 otherwise, indicating that the AL distribution can simultaneously deal with the symmetric and asymmetric scenarios.

According to (4) and (5), the following equation holds:

$$y_{ij} \sim \mathrm{AL}_j\left(y_{ij}\middle|x_{ij}, \lambda_j, \kappa_j\right) \quad \forall j = 1, 2, \ldots, B \tag{7}$$

and the log-likelihood function is

$$\begin{aligned}\ell &= \sum_{i=1}^{MN}\sum_{j=1}^{B}\log \mathrm{AL}_j\left(x_{ij}, \lambda_j, \kappa_j\right) \\ &= \sum_{i=1}^{MN}\sum_{j=1}^{B}\log\left[\lambda_j\kappa_j\left(1-\kappa_j\right)\right] - \left|y_{ij}-x_{ij}\right|\lambda_j\eta_{ij}.\end{aligned} \tag{8}$$

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B. Solving Algorithm

Now, let us move on to the model optimization. The parameters in BALMF can be iteratively updated by maximizing (8) according to the MLE principle.

1) Updating λ_j: To start with, let us focus on λ_j. The objective function with regard to (w.r.t.) λ_j is

$$\ell\left(\lambda_j\right) = \sum_{i=1}^{MN} \left(\log \lambda_j - \lambda_j \eta_{ij} \left|y_{ij} - x_{ij}\right|\right). \tag{9}$$

It is easy to write the gradient as follows:

$$\frac{\partial \ell\left(\lambda_j\right)}{\partial \lambda_j} = \sum_{i=1}^{MN} \left(\frac{1}{\lambda_j} - \eta_{ij} \left|y_{ij} - x_{ij}\right|\right). \tag{10}$$

Let it be zero and we have the following updating rule:

$$\lambda_j = \frac{MN}{\sum_{i=1}^{MN} \eta_{ij} \left|y_{ij} - x_{ij}\right|}. \tag{11}$$

2) Updating κ_j: The objective function with respect to κ_j is

$$\ell\left(\kappa_j\right) = \sum_{i=1}^{MN} \left[\log \kappa_j + \log\left(1 - \kappa_j\right) - \lambda_j \eta_{ij} \left|y_{ij} - x_{ij}\right|\right] \tag{12}$$

and the gradient with respect to κ_j is given by

$$\frac{\partial \ell\left(\kappa_j\right)}{\partial \kappa_j} = \sum_{i=1}^{MN} \left(\frac{1}{\kappa_j} - \frac{1}{1 - \kappa_j} - \lambda_j \left|y_{ij} - x_{ij}\right| \frac{\partial \eta_{ij}}{\partial \kappa_j}\right). \tag{13}$$

Note that $\partial \eta_{ij} / \partial \kappa_j = \mathbb{I}(y_{ij} - x_{ij} \geqslant 0) - \mathbb{I}(y_{ij} - x_{ij} < 0) = \operatorname{sign}(y_{ij} - x_{ij})$, where $\operatorname{sign}(y_{ij} - x_{ij})$ represents the sign function for $y_{ij} - x_{ij}$, taking 1 if $y_{ij} - x_{ij} \geqslant 0$ and -1 otherwise, so $|y_{ij} - x_{ij}| \cdot \partial \eta_{ij} / \partial \kappa_j = |y_{ij} - x_{ij}| \cdot \operatorname{sign}(y_{ij} - x_{ij}) = y_{ij} - x_{ij}$. Equation (13) then can be rewritten as

$$\frac{\partial \ell\left(\kappa_j\right)}{\partial \kappa_j} = \sum_{i=1}^{MN} \left(\frac{1}{\kappa_j} - \frac{1}{1 - \kappa_j} - \lambda_j \left(y_{ij} - x_{ij}\right)\right). \tag{14}$$

By letting $(\partial \ell(\kappa_j)) / (\partial \kappa_j) = 0$, we can get

$$\xi_j \kappa_j^2 - \left(\xi_j + 2MN\right) \kappa_j + MN = 0 \tag{15}$$

where

$$\xi_j = \lambda_j \sum_{i=1}^{MN} \left(y_{ij} - x_{ij}\right). \tag{16}$$

This two-order equation only has one root satisfying the constraint $0 < \kappa < 1$ (see details in *Supplementary Material*), i.e.,

$$\kappa_j = \frac{\xi_j + 2MN - \sqrt{\xi_j^2 + 4M^2N^2}}{2\xi_j}. \tag{17}$$

3) Updating $\boldsymbol{U}$ and $\boldsymbol{V}$: As for factor matrices, it corresponds to the following optimization problem:

$$\min_{\boldsymbol{U},\boldsymbol{V}} \sum_{i=1}^{MN} \sum_{j=1}^{B} \lambda_j \eta_{ij} \left|y_{ij} - \boldsymbol{u}_{i,:} \boldsymbol{v}_{j,:}^{\mathrm{T}}\right|. \tag{18}$$

It is difficult to solve the above issue, since there is a tiresome ℓ_1-norm loss. Let the (i, j) entry of $\boldsymbol{W}$ be $w_{ij} = \lambda_j \eta_{ij}$, and we rearrange this problem in the matrix form as follows:

$$\min_{\boldsymbol{U},\boldsymbol{V}} \left\| \boldsymbol{W} \odot \left(\boldsymbol{Y} - \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}\right)\right\|_1 \tag{19}$$

which is equivalent to

$$\min_{\boldsymbol{X},\boldsymbol{Z},\boldsymbol{U},\boldsymbol{V}} \|\boldsymbol{Z}\|_1, \quad \text{s.t. } \boldsymbol{Z} = \boldsymbol{W} \odot (\boldsymbol{Y} - \boldsymbol{X}), \boldsymbol{X} = \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} \tag{20}$$

where $\boldsymbol{Z} \in \mathbb{R}^{MN \times B}$ and $\boldsymbol{X} \in \mathbb{R}^{MN \times B}$ are two auxiliary variables, and the symbol $\odot$ denotes the elementwise product. The half quadratic splitting (HQS) algorithm is used to solve (20). First, we convert the constraint optimization problem into an unconstrained optimization problem shown as follows:

$$\min_{\boldsymbol{X},\boldsymbol{Z},\boldsymbol{U},\boldsymbol{V}} \|\boldsymbol{Z}\|_1 + \frac{\rho_1}{2} \|\boldsymbol{Z} - \boldsymbol{W} \odot (\boldsymbol{Y} - \boldsymbol{X})\|_2^2 + \frac{\rho_2}{2} \left\|\boldsymbol{X} - \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}\right\|_2^2 \tag{21}$$

where ρ_1 and ρ_2 are two nonnegative penalty parameters. Then, we iteratively optimize (21) with respect to one variable while fixing others, and it will be divided into three subproblems.

For the auxiliary variable $\boldsymbol{Z}$, the objective function is rewritten as follows:

$$\min_{\boldsymbol{Z}} \|\boldsymbol{Z}\|_1 + \frac{\rho_1}{2} \|\boldsymbol{Z} - \boldsymbol{W} \odot (\boldsymbol{Y} - \boldsymbol{X})\|_2^2. \tag{22}$$

This is an ℓ_1-norm penalized least-square problem whose solution is given by

$$\boldsymbol{Z} = S_{1/\rho_1} \left(\boldsymbol{W} \odot (\boldsymbol{Y} - \boldsymbol{X})\right) \tag{23}$$

where $S_\alpha(x) = \operatorname{sign}(x) \max(|x| - \alpha, 0)$ denotes the soft-thresholding function.

For the auxiliary variable $\boldsymbol{X}$, the objective function is rewritten as follows:

$$\min_{\boldsymbol{X}} \frac{\rho_1}{2} \|\boldsymbol{Z} - \boldsymbol{W} \odot (\boldsymbol{Y} - \boldsymbol{X})\|_2^2 + \frac{\rho_2}{2} \left\|\boldsymbol{X} - \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}\right\|_2^2. \tag{24}$$

With some simple calculus operations, we can get the solution

$$\boldsymbol{X} = \frac{\boldsymbol{W}^2 \odot \boldsymbol{Y} - \boldsymbol{W} \odot \boldsymbol{Z} + \rho_2 \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}} / \rho_1}{\boldsymbol{W}^2 + \rho_2 / \rho_1}. \tag{25}$$

Finally, the objective function with respect to factor matrices $\boldsymbol{U}$ and $\boldsymbol{V}$ is $\min_{\boldsymbol{U} \in \mathbb{R}^{MN \times R}, \boldsymbol{V} \in \mathbb{R}^{B \times R}} \|\boldsymbol{X} - \boldsymbol{U}\boldsymbol{V}^{\mathrm{T}}\|_2^2$. Its solution can be written as follows:

$$[\boldsymbol{U}, \boldsymbol{V}] = \mathrm{SVD}_R(\boldsymbol{X}) \tag{26}$$

where $\mathrm{SVD}_R(\cdot)$ is the truncated SVD with rank R.

Algorithm 1 summarizes the HQS algorithm for updating $\boldsymbol{U}$ and $\boldsymbol{V}$. Note that the scaling parameter γ is a constant larger than 1 making ρ_1 and ρ_2 increase in each step to promote convergence. According to the analysis reported in Section IV-D,

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Algorithm 1 Solving (21)

Input: The input matrix $\boldsymbol{Y}$, the weight matrix $\boldsymbol{W}$, the maximum iteration *InMaxIter*, tolerance *InTol*, scaling parameter $\gamma > 1$.

Output: $\boldsymbol{U}$,$\boldsymbol{V}$

$iter = 1$, $\boldsymbol{X}^{\text{old}} = \boldsymbol{U}\boldsymbol{V}^{\text{T}}$;
while *not converged* **do**
  Update $\boldsymbol{Z}$ by (23) ;
  Update $\boldsymbol{X}$ by (25) ;
  Update $\boldsymbol{U}$ and $\boldsymbol{V}$ by (26) ;
  $iter = iter + 1$, $\boldsymbol{X}^{\text{new}} = \boldsymbol{U}\boldsymbol{V}^{\text{T}}$;
  if $\frac{\|\boldsymbol{X}^{\text{new}} - \boldsymbol{X}^{\text{old}}\|_2^2}{\|\boldsymbol{X}^{\text{old}}\|_2^2} < InTol$ *or* $iter > InMaxIter$ **then**
    break;
  else
    $\boldsymbol{X}^{\text{old}} = \boldsymbol{X}^{\text{new}}$; $\rho_1 = \gamma\rho_1$, $\rho_2 = \gamma\rho_2$.
  end
end

Algorithm 2 BALMF

Input: The input matrix $\boldsymbol{Y}$, the maximum iteration *MaxIter*, tolerance *Tol*.

Output: $\boldsymbol{U}$,$\boldsymbol{V}$

$iter = 1$, $\boldsymbol{X}^{\text{old}} = \boldsymbol{U}\boldsymbol{V}^{\text{T}}$;
while *not converged* **do**
  Update λ_j by (11) ($j = 1, \ldots, B$);
  Update ξ_j by (16) ($j = 1, \ldots, B$);
  Update κ_j by (17) ($j = 1, \ldots, B$);
  Update $\boldsymbol{U}$ and $\boldsymbol{V}$ by Algorithm 1 with iteration number $InMaxIter$ and tolerance $InTol$;
  $iter = iter + 1$, $\boldsymbol{X}^{\text{new}} = \boldsymbol{U}\boldsymbol{V}^{\text{T}}$;
  if $\frac{\|\boldsymbol{X}^{\text{new}} - \boldsymbol{X}^{\text{old}}\|_2^2}{\|\boldsymbol{X}^{\text{old}}\|_2^2} < Tol$ *or* $iter > MaxIter$ **then**
    break;
  else
    $\boldsymbol{X}^{\text{old}} = \boldsymbol{X}^{\text{new}}$;
  end
end

we set $\gamma = 1.5$, $\rho_1 = 10$ and $\rho_2 = \rho_1 * \text{mean}(\boldsymbol{W}^2)/2$. This algorithm stops when the iteration reaches the maximal value InMaxIter or $(\|\boldsymbol{X}^{\text{new}} - \boldsymbol{X}^{\text{old}}\|_2^2)/(\|\boldsymbol{X}^{\text{old}}\|_2^2)$ is smaller than a threshold InTol.

Now, we have demonstrated how to solve the BALMF model. Algorithm 2 summarizes the workflow. It iteratively updates $\lambda_j, \kappa_j, \boldsymbol{U}$ and $\boldsymbol{V}$ until the iteration reaches the maximal value MaxIter or $(\|\boldsymbol{X}^{\text{new}} - \boldsymbol{X}^{\text{old}}\|_2^2)/(\|\boldsymbol{X}^{\text{old}}\|_2^2)$ is smaller than a threshold Tol. In this article, we consistently set MaxIter = 20, Tol = 10^{-4}, InMaxIter = 30 and InTol = 10^{-4} throughout all the experiments.

C. Computational Complexity

In Algorithm 1, the computational complexities for updating $\boldsymbol{Z}$ and $\boldsymbol{X}$ are $O(MNB)$ and $O(MNB)$, respectively. If using bidiagonalization and the QR algorithm to carry out SVD, the computational complexity will be $O(MNB^2)$. Therefore, the computational complexity of Algorithm 1 is $O(n_{\text{in}}(MNB + MNB^2))$, where n_{in} is the number of iterations for Algorithm 1. In Algorithm 2, the computational complexities of updating λ, η, and κ are $O(B)$, $O(B)$, and $O(B)$, respectively. In summary, the computational complexity of our proposed method thus is $O(n_{\text{out}}(n_{\text{in}}(MNB + MNB^2) + B))$, where n_{out} is the number of iterations for Algorithm 2.

IV. Experiments

The synthetic and real-world data experiments are carried out on a computer with an Intel Core i7-12700K CPU at 3.6 GHz and 32-GB RAM. The SOTA methods are selected as competing algorithms. They are as follows.

1) *Low-Rank-Based Methods:* Noise-adjusted iterative low-rank matrix approximation (NAILRMA) [53], hyperspectral restoration (HyRes) [54], fast hyperspectral denoising (FastHyDe) [55], LRMR [9], and nonconvex low-rank matrix approximation (NonLRMA) [19].
2) *TV Regularized Low-Rank Methods:* TV regularized low-rank method (LRTV) [14] and weighted group sparsity-regularized low-rank tensor decomposition (LRTDGS) [56].
3) *Non-Local-Based Methods:* Nonlocal meets global (NGMeet) [39].
4) *Noise Modeling Methods:* NMoG [13].
5) *Deep-Learning-Based Method:* CNN based HSI denoising method (HSIDCNN) [45].
6) *Hybrid Methods of Low-Rank Models and Neural Networks:* Fast and parameter-free hyperspectral image mixed noise removal method (FastHyMix) [57] and HSI mixed noise removal method using subspace representation and deep CNN image prior (HySuDeep) [58].

The evaluation metrics include peak signal-to-noise ratio (PSNR), structural similarity (SSIM), feature similarity (FSIM), erreur relative global adimensionnelle de synthèse (ERGAS), and spectral angle mapper (SAM).

A. Synthetic Data Experiments

In the simulated experiments, a subimage with 80 bands and 200 × 200 pixels is cropped from the Pavia city center (PCC) dataset which was captured by the Reflective Optics System Imaging Spectrometer (ROSIS-03), and a subimage with 191 bands and 256 × 256 pixels is cropped from the Washington DC Mall (WDCM) dataset which was captured by the Hyperspectral Digital Imagery Collection Experiment (HYDICE).

Different kinds of noise types are simulated to degrade PCC and WDCM. Eight cases are considered, including the mixture of Gaussian noise with different standard deviations σ, impulse noise with different intensities p, and stripes/deadlines with different proportions q across different bands of the implemented HSIs.

Case 1 (i.i.d Gaussian Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.05$.

Case 2 (Non-i.i.d Gaussian Noise): Each band is corrupted by Gaussian noise with σ randomly sampled from 0.01 to 0.05.

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TABLE I

METRICS ON THE PCC DATASET. ↑/↓ INDICATES THAT BETTER RESULTS ARE WITH LARGER/SMALLER METRIC VALUES. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND ITALIC, RESPECTIVELY

Cases	Metrics	Noisy	NAIRLMA	HyRes	FastHyDe	LRMR	NonLRMA	NGMeet	LRTV	LRTDGS	NMoG	HSIDCNN	FastHyMix	HySuDeep	BALMF
1	PSNR↑	26.29	39.43	39.33	*39.91*	37.25	33.61	**41.17**	39.46	38.54	39.01	37.34	39.92	39.93	38.13
	SSIM↑	0.7295	0.9796	0.9808	*0.9837*	0.9731	0.9484	**0.9872**	0.9815	0.9773	0.9786	0.9704	0.9838	0.9837	0.9722
	FSIM↑	0.8772	0.9893	0.9890	*0.9903*	0.9825	0.9639	**0.9925**	0.9890	0.9871	0.9885	0.9811	0.9903	0.9900	0.9852
	ERGAS↓	177.71	39.85	39.98	*37.28*	52.43	84.91	**32.26**	39.84	43.99	41.76	52.71	37.27	37.39	47.02
	SAM↓	0.3043	0.0630	0.0630	*0.0583*	0.0816	0.1105	**0.0481**	0.0676	0.0696	0.0665	0.0887	0.0584	0.0621	0.0809
2	PSNR↑	31.31	*43.66*	**44.34**	42.28	39.85	36.81	41.49	42.48	40.56	43.28	39.57	43.51	43.32	41.35
	SSIM↑	0.8612	0.9921	**0.9934**	0.9906	0.9845	0.9697	0.9823	0.9913	0.9858	*0.9925*	0.9805	0.9931	0.9928	0.9882
	FSIM↑	0.9346	*0.9956*	**0.9962**	0.9941	0.9897	0.9780	0.9912	0.9945	0.9923	0.9952	0.9872	0.9954	0.9952	0.9929
	ERGAS↓	117.36	*24.85*	**22.84**	28.75	39.65	66.05	38.59	28.29	34.96	26.02	42.63	25.31	25.86	32.55
	SAM↓	0.2220	*0.0461*	**0.0440**	0.0511	0.0670	0.0904	0.0707	0.0523	0.0589	0.0495	0.0759	0.0476	0.0495	0.0593
3	PSNR↑	14.52	27.21	27.54	28.20	36.92	25.87	28.28	37.66	41.81	44.24	38.69	40.35	*44.42*	**44.44**
	SSIM↑	0.2742	0.8540	0.8675	0.8993	0.9712	0.8330	0.9023	0.9792	0.9899	**0.9952**	0.9797	0.9895	0.9949	*0.9950*
	FSIM↑	0.7189	0.9288	0.9318	0.9498	0.9815	0.9102	0.9519	0.9867	0.9947	*0.9965*	0.9877	0.9938	0.9965	**0.9966**
	ERGAS↓	643.20	149.76	144.26	134.23	54.35	174.46	133.33	48.22	30.38	24.12	42.76	40.82	*23.39*	**23.16**
	SAM↓	0.6911	0.1286	0.1177	0.1067	0.0829	0.1506	0.1022	0.0526	0.0534	**0.0435**	0.0794	0.0576	0.0448	*0.0445*
4	PSNR↑	14.39	26.25	26.30	27.24	32.30	25.27	27.09	36.63	35.85	42.31	37.68	37.09	*43.22*	**43.72**
	SSIM↑	0.2559	0.8252	0.8164	0.8745	0.9155	0.8146	0.8588	0.9534	0.9498	0.9902	0.9738	0.9604	*0.9935*	**0.9946**
	FSIM↑	0.6978	0.9104	0.9046	0.9340	0.9513	0.8993	0.9314	0.9732	0.9721	0.9945	0.9839	0.9820	*0.9958*	**0.9964**
	ERGAS↓	681.91	185.89	190.28	165.39	167.65	200.50	180.44	113.00	112.74	61.02	50.55	159.59	*28.91*	**25.55**
	SAM↓	0.7329	0.2045	0.2092	0.1801	0.1925	0.2149	0.1926	0.1315	0.1348	0.0702	0.0879	0.1567	*0.0512*	**0.0465**
5	PSNR↑	14.35	26.15	26.23	27.25	31.78	25.24	27.06	37.07	37.58	41.88	37.30	35.87	*43.01*	**43.61**
	SSIM↑	0.2518	0.8275	0.8151	0.8749	0.9111	0.8140	0.8586	0.9593	0.9680	0.9831	0.9715	0.9558	*0.9928*	**0.9945**
	FSIM↑	0.6922	0.9108	0.9045	0.9347	0.9483	0.8989	0.9307	0.9758	0.9820	0.9917	0.9822	0.9775	*0.9954*	**0.9963**
	ERGAS↓	682.71	184.30	189.00	162.79	167.13	199.09	176.91	106.25	82.75	100.89	52.59	165.54	*33.41*	**25.89**
	SAM↓	0.7323	0.2008	0.2083	0.1758	0.1938	0.2103	0.1868	0.1295	0.1092	0.1132	0.0916	0.1636	*0.0581*	**0.0469**
6	PSNR↑	12.75	24.10	23.97	24.69	29.41	23.74	24.84	35.12	34.39	*38.01*	35.44	28.51	37.44	**41.03**
	SSIM↑	0.1647	0.7653	0.7386	0.7942	0.8740	0.7616	0.8022	0.9388	0.9331	0.9606	0.9580	0.8930	*0.9755*	**0.9908**
	FSIM↑	0.6112	0.8824	0.8680	0.8961	0.9287	0.8778	0.9024	0.9656	0.9633	0.9832	0.9736	0.9459	*0.9867*	**0.9949**
	ERGAS↓	802.48	221.11	228.23	208.28	205.99	228.25	206.69	134.34	136.67	163.00	67.73	266.50	*64.75*	**39.35**
	SAM↓	0.8064	0.2309	0.2411	0.2204	0.2321	0.2339	0.2146	0.1619	0.1664	0.1602	0.1099	0.2987	*0.0820*	**0.0573**
7	PSNR↑	11.64	22.67	22.24	23.45	27.34	22.68	23.33	32.44	31.78	32.01	*33.71*	25.93	33.40	**37.70**
	SSIM↑	0.1165	0.7067	0.6462	0.7550	0.8358	0.7080	0.7442	0.9204	0.9122	0.9303	0.9418	0.8419	*0.9543*	**0.9832**
	FSIM↑	0.5531	0.8575	0.8246	0.8733	0.9105	0.8578	0.8724	0.9570	0.9532	0.9695	0.9637	0.9135	*0.9758*	**0.9919**
	ERGAS↓	896.68	253.35	268.18	232.75	246.69	252.76	237.06	156.58	162.98	218.92	*82.35*	302.10	95.80	**59.38**
	SAM↓	0.8556	0.2546	0.2739	0.2372	0.2658	0.2502	0.2384	0.1801	0.1903	0.1914	0.1285	0.3595	*0.1073*	**0.0707**
8	PSNR↑	10.83	21.74	21.24	22.22	25.99	21.91	22.32	29.92	29.46	28.83	*30.16*	22.65	29.69	**31.05**
	SSIM↑	0.0876	0.6485	0.5813	0.6677	0.8023	0.6588	0.6787	0.8857	0.8752	0.8963	0.9015	0.7451	*0.9050*	**0.9181**
	FSIM↑	0.5097	0.8337	0.7973	0.8355	0.8932	0.8377	0.8385	0.9408	0.9354	*0.9562*	0.9512	0.8753	0.9502	**0.9609**
	ERGAS↓	966.49	277.55	293.58	264.95	273.98	273.62	262.16	199.96	201.85	273.57	196.49	307.18	*172.07*	**169.66**
	SAM↓	0.8859	0.2701	0.2910	0.2625	0.2878	0.2611	0.2539	0.2152	0.2248	0.2308	0.1938	0.3455	*0.1804*	**0.1742**

Case 3 (Gaussian Noise + Impulse Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.1$.

Case 4 (Gaussian Noise + Impulse Noise + Deadlines): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.1$. In addition, 20% bands are degraded by deadlines with $q = 35$.

Case 5 (Mixture Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.1$. In addition, 20% bands are degraded by deadlines with $q = 35$, and 20% bands are degraded by stripes with $q = 35$.

Case 6 (Mixture Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.15$. In addition, 30% bands are degraded by deadlines with $q = 35$, and 30% bands are degraded by stripes with $q = 35$.

Case 7 (Mixture Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.2$. In addition, 40% bands are degraded by deadlines with $q = 35$, and 40% bands are degraded by stripes with $q = 35$.

Case 8 (Mixture Noise): The HSI cube is corrupted by Gaussian noise with $\sigma = 0.01$ and impulse noise with $p = 0.25$. In addition, 50% bands are degraded by deadlines with $q = 35$, and 50% bands are degraded by stripes with $q = 35$.

For compared methods, the hyperparameters are tuned by grid search to possibly guarantee their optimal performance. HSIDCNN is trained over 100 epochs on 12 800 training pairs cropped from the PCC dataset, where simulated noise is generated by the aforementioned eight cases. Tables I and II report the quantitative metrics for the eight cases on the PCC and WDCM datasets, respectively. The best and second best results are highlighted in bold and italic typefaces, respectively. From this table, the following observations can be obtained.

1) NGMeet, an SOTA Gaussian denoiser, achieves the best performance in the Gaussian noise case. When the Gaussian noise intensity varies across bands in Case 2, HyRes and NAIRLMA have better performance. Although our proposed BALMF is not specifically designed for Gaussian noise, it can still obtain comparable results with most of the competing methods in this simple ideal case.
2) Cases 3–8 contain the mixture of different noise types and are more similar with real noise. BALMF takes the first or second place in these cases. This can be rationally explained by that BALMF can finely model the heavy-tailness and asymmetry for each band, and the elaborate noise modeling is relatively more suitable for real HSI noise.
3) HSIDCNN achieves the second best results in Cases 7 and 8, but performs badly in other cases, which attributes to relatively weak generalization capability of

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TABLE II

METRICS ON THE WDCM DATASET. ↑/↓ INDICATES THAT BETTER RESULTS ARE WITH LARGER/SMALLER METRIC VALUES. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND ITALIC, RESPECTIVELY

Cases	Metrics	Noisy	NAIRLMA	HyRes	FastHyDe	LRMR	NonLRMA	NGMeet	LRTV	LRTDGS	NMoG	HSIDCNN	FastHyMix	HySuDeep	BALMF
1	PSNR↑	26.47	*40.08*	39.86	36.79	36.85	35.80	**40.56**	36.75	36.11	36.68	36.00	36.82	36.98	36.49
	SSIM↑	0.7205	0.9812	*0.9815*	0.9760	0.9701	0.9672	**0.9848**	0.9763	0.9709	0.9748	0.9602	0.9762	0.9774	0.9739
	FSIM↑	0.8821	*0.9906*	0.9905	0.9842	0.9812	0.9782	**0.9922**	0.9838	0.9821	0.9839	0.9771	0.9843	0.9851	0.9834
	ERGAS↓	195.98	*41.56*	42.59	64.56	60.42	68.68	**39.18**	65.84	69.24	65.19	66.51	64.17	63.78	67.51
	SAM↓	0.3165	0.0753	*0.0749*	0.0999	0.1020	0.1103	**0.0681**	0.0966	0.1019	0.1025	0.1036	0.1003	0.0995	0.1023
2	PSNR↑	31.59	*44.21*	**44.59**	37.53	38.96	39.99	42.10	37.47	36.56	37.84	37.97	37.77	37.73	37.26
	SSIM↑	0.8629	*0.9922*	**0.9932**	0.9807	0.9810	0.9852	0.9903	0.9809	0.9748	0.9820	0.9741	0.9822	0.9820	0.9804
	FSIM↑	0.9409	*0.9959*	**0.9963**	0.9865	0.9874	0.9899	0.9942	0.9861	0.9841	0.9871	0.9847	0.9874	0.9876	0.9860
	ERGAS↓	125.15	*26.63*	**25.62**	61.33	48.74	45.65	34.03	62.61	66.75	60.71	54.49	60.01	60.65	63.96
	SAM↓	0.2226	*0.0517*	**0.0483**	0.0938	0.0864	0.0831	0.0619	0.0910	0.0976	0.0921	0.0862	0.0922	0.0938	0.0927
3	PSNR↑	14.29	26.56	26.72	27.34	36.97	23.14	27.33	**41.63**	39.21	41.19	35.93	38.99	37.94	*41.25*
	SSIM↑	0.3004	0.8247	0.8369	0.8699	0.9736	0.6955	0.8673	**0.9922**	0.9846	0.9922	0.9605	0.9892	0.9829	*0.9922*
	FSIM↑	0.7301	0.9345	0.9353	0.9527	0.9842	0.8534	0.9526	**0.9949**	0.9918	0.9947	0.9779	0.9928	0.9883	*0.9949*
	ERGAS↓	712.44	171.73	168.71	158.20	57.38	254.21	158.29	**36.78**	45.50	38.97	65.74	47.20	60.46	*37.64*
	SAM↓	0.7118	0.1590	0.1479	0.1339	0.0864	0.2530	0.1357	**0.0540**	0.0626	0.0548	0.0926	0.0581	0.0942	*0.0548*
4	PSNR↑	14.18	26.59	26.50	27.26	32.99	24.13	27.08	36.92	36.23	36.77	34.69	33.53	*36.94*	**37.24**
	SSIM↑	0.2676	0.8423	0.8276	0.8762	0.9315	0.7476	0.8602	**0.9803**	0.9744	0.9779	0.9483	0.9472	0.9790	*0.9801*
	FSIM↑	0.6986	0.9310	0.9271	0.9486	0.9593	0.8734	0.9444	0.9857	0.9840	0.9848	0.9707	0.9772	*0.9862*	**0.9865**
	ERGAS↓	762.59	192.88	199.00	178.37	154.15	247.02	188.55	*68.59*	70.62	76.59	78.12	75.54	73.54	**68.56**
	SAM↓	0.7868	0.2270	0.2303	0.2053	0.1867	0.2883	0.2171	*0.0928*	0.0985	0.0941	0.1043	0.1311	0.1079	**0.0922**
5	PSNR↑	14.12	26.30	26.23	27.00	32.32	23.94	26.81	36.84	36.16	36.78	34.38	34.30	*37.07*	**37.23**
	SSIM↑	0.2618	0.8364	0.8201	0.8705	0.9256	0.7428	0.8521	*0.9801*	0.9743	0.9780	0.9448	0.9599	0.9794	**0.9802**
	FSIM↑	0.6916	0.9286	0.9241	0.9467	0.9555	0.8720	0.9419	0.9858	0.9840	0.9849	0.9684	0.9784	*0.9861*	**0.9867**
	ERGAS↓	764.93	199.74	205.81	183.64	160.64	251.08	193.23	*69.78*	71.30	77.09	80.80	147.67	71.98	**68.99**
	SAM↓	0.7920	0.2348	0.2409	0.2107	0.1975	0.2935	0.2228	0.0948	0.0994	*0.0944*	0.1094	0.1643	0.1067	**0.0940**
6	PSNR↑	12.56	24.07	23.92	24.87	30.23	22.79	24.70	35.51	35.46	*36.07*	32.75	32.09	35.16	**36.12**
	SSIM↑	0.1759	0.7723	0.7484	0.8158	0.8963	0.7038	0.7996	*0.9756*	0.9714	0.9712	0.9247	0.9411	0.9709	**0.9763**
	FSIM↑	0.6185	0.9033	0.8943	0.9259	0.9397	0.8585	0.9223	0.9836	0.9827	0.9825	0.9572	0.9704	*0.9842*	**0.9852**
	ERGAS↓	883.66	239.87	246.44	220.32	193.30	272.88	226.19	82.78	**77.77**	91.76	97.50	179.96	90.31	*80.05*
	SAM↓	0.8495	0.2619	0.2680	0.2340	0.2180	0.3026	0.2418	0.1036	*0.1033*	0.1065	0.1250	0.1814	0.1168	**0.0987**
7	PSNR↑	11.40	21.95	21.80	22.96	28.25	21.57	22.74	34.00	34.33	**35.23**	31.37	28.81	32.72	*35.06*
	SSIM↑	0.1259	0.6907	0.6530	0.7561	0.8631	0.6622	0.7314	*0.9664*	0.9653	0.9629	0.9019	0.8980	0.9524	**0.9709**
	FSIM↑	0.5628	0.8678	0.8541	0.9021	0.9221	0.8452	0.8933	0.9798	0.9802	*0.9809*	0.9444	0.9485	0.9766	**0.9847**
	ERGAS↓	974.33	293.85	298.96	263.07	229.09	304.56	269.70	105.33	**92.46**	116.26	116.67	220.05	123.97	*93.77*
	SAM↓	0.9001	0.3166	0.3218	0.2776	0.2600	0.3299	0.2852	0.1245	*0.1180*	0.1267	0.1481	0.2612	0.1383	**0.1101**
8	PSNR↑	10.51	20.17	20.16	21.08	26.18	20.50	21.13	*30.20*	28.22	29.54	30.00	25.50	29.48	**32.47**
	SSIM↑	0.0937	0.5949	0.5639	0.6606	0.8156	0.6153	0.6510	*0.9287*	0.8889	0.9167	0.8727	0.8321	0.9147	**0.9486**
	FSIM↑	0.5189	0.8251	0.8168	0.8569	0.8961	0.8277	0.8547	*0.9593*	0.9353	0.9535	0.9287	0.9136	0.9572	**0.9782**
	ERGAS↓	1050.24	351.50	351.04	318.57	281.42	339.37	316.74	169.31	206.20	244.37	*137.91*	275.53	187.58	**136.93**
	SAM↓	0.9463	0.3770	0.3786	0.3381	0.3063	0.3599	0.3333	0.1755	0.2346	*0.1663*	0.1701	0.3129	0.1699	**0.1336**

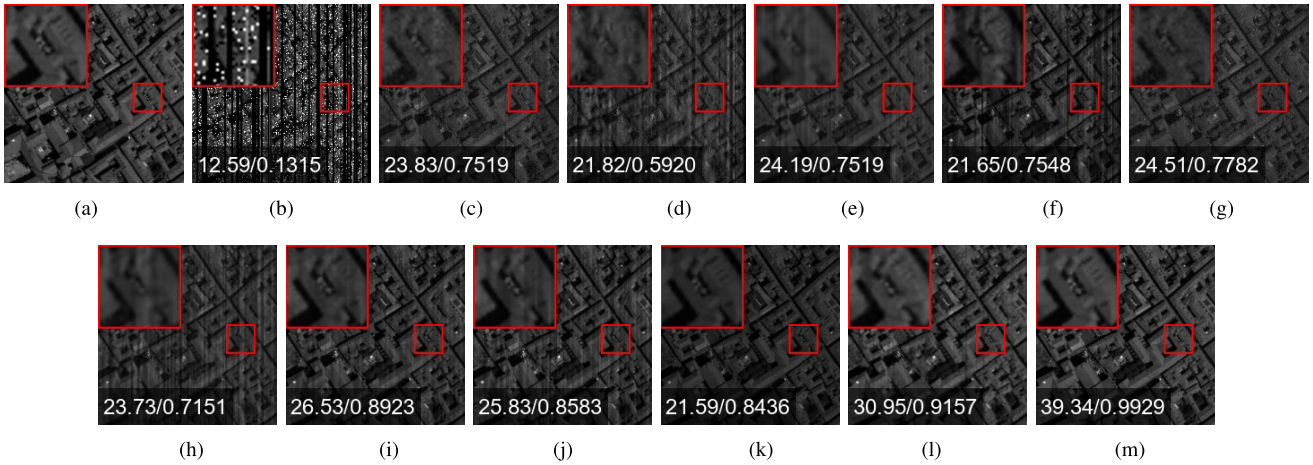


Fig. 3. Denoised HSI images obtained by different competing methods on the PCC dataset (Band 80) in case 6. The PSNR/SSIM value is shown in the bottom left corner. (a) Ground truth. (b) Observation. (c) NAIRLMA. (d) HyRes. (e) FastHyDe. (f) LRMR. (g) NonLRMA. (h) NGMeet. (i) LRTV. (j) LRTDGS. (k) NMoG. (l) HSIDCNN. (m) BALMF.

the deep-learning-based method when compared with the prior-model-based methods. That is, HSIDCNN is good at removing a specific pattern of noise, but it inclines to lose efficiency when noise pattern dramatically varies. FastHyMix and HySuDeep are two representative hybrid methods, where pretrained neural networks serving as denoising prior are inserted into the low-rank models. FastHyMix outperforms HSIDCNN in Cases 1–3, and HySuDeep outperforms HSIDCNN in Cases 3–6, but they are worse than our BALMF, which implies that combining neural networks and low-rank models is able to ameliorate generalization of the deep learning methods to some degree, but still suffers similar problems.

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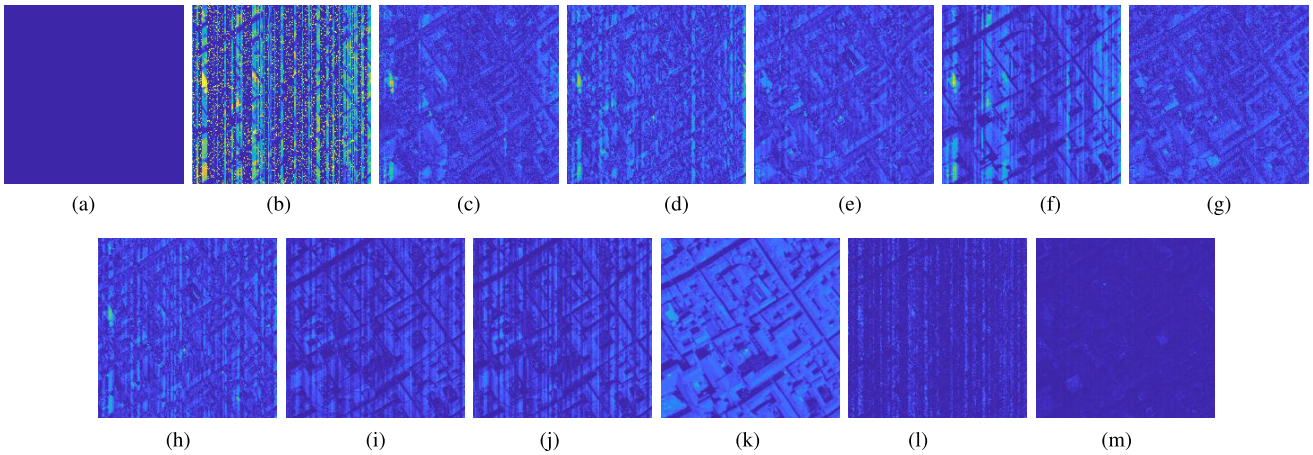


Fig. 4. Error maps corresponding to Fig. 3. From red to yellow, the error goes larger. (a) Ground truth. (b) Observation. (c) NAIRLMA. (d) HyRes. (e) FastHyDe. (f) LRMR. (g) NonLRMA. (h) NGMeet. (i) LRTV. (j) LRTDGS. (k) NMoG. (l) HSIDCNN. (m) BALMF.

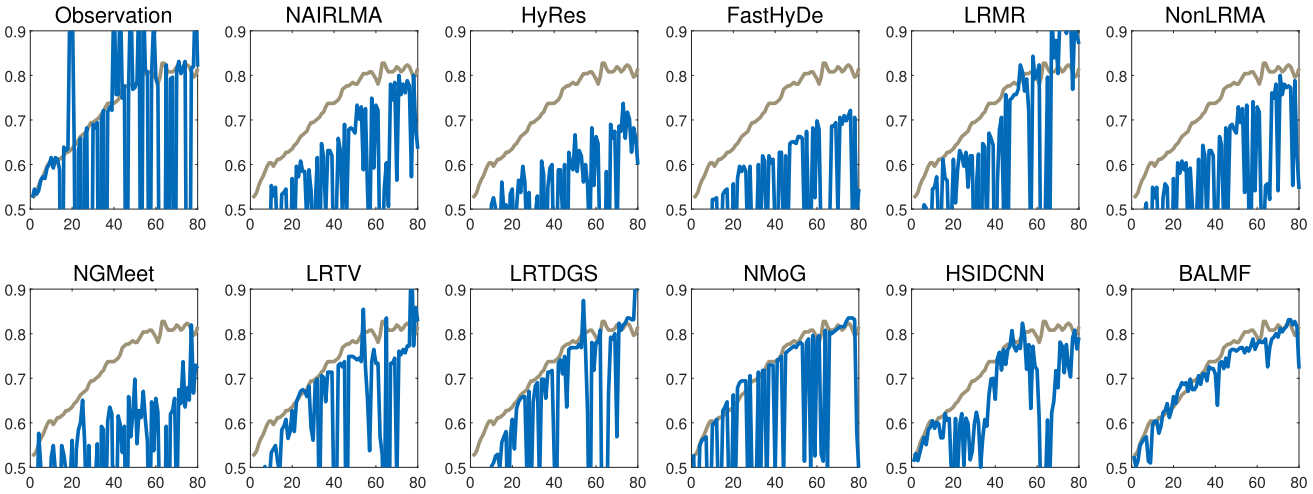


Fig. 5. Spectral curves of all the competing methods at pixel (164, 43) on the PCC dataset in Case 7. Brown and blue lines are the ground truth and denoised curve, respectively. The horizontal axis is the band number, and the vertical axis is the digital value.

Band 80 of the PCC dataset in Case 6 is selected for visual comparison of all the methods, which is demonstrated in Fig. 3. A local area shown in the top left corner is amplified to exhibit the detailed information. Compared with the clean image in Fig. 3(a), the observed noisy image in Fig. 3(b) is obviously degraded by the mixture of Gaussian noise, impulse noise, deadlines, and stripes. Denoised images are given in Figs. 3(c)–(m) and 4 visualizes the corresponding error maps. There is no doubt that all the methods could remove noise less or more. Many stripes and deadlines still remain in the images denoised by HyRes, FastHyDe, LRMR, NGMeet, LRTV, and LRTDGS. As for NAIRLMA and NonLRMA, they remains Gaussian-like noise. NMoG can remove most of the noise, but there is a great gap between the image restored by NMoG and the ground-truth image in terms of luminance. As a result, NMoG may lead to spectral distortion. Comparatively, our BALMF outperforms all the other competing methods, since it is capable of recovering cleaner structures and preserving more local details.

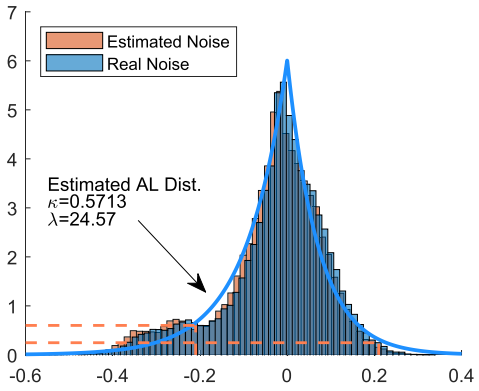


Fig. 6. Noise histogram on the PCC dataset.

In addition to quantitative metrics, Fig. 5 displays the spectral curves at pixel (164, 43) on the PCC dataset in Case 7. The ground-truth spectrum is marked by brown, and the denoised one is represented by red. It is clear that the observed

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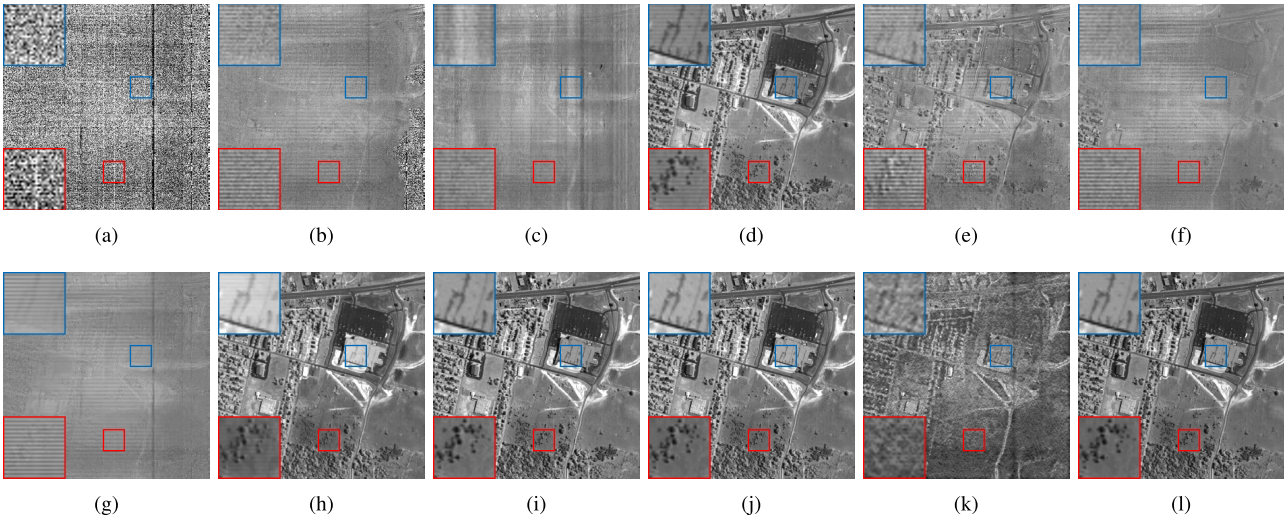


Fig. 7. Denoised HSIs obtained by all the competing methods on band 209 of the Urban dataset. (a) Observation. (b) NAIRLMA. (c) HyRes. (d) FastHyDe. (e) LRMR. (f) NonLRMA. (g) NGMeet. (h) LRTV. (i) LRTDGS. (j) NMoG. (k) HSIDCNN. (l) BALMF.

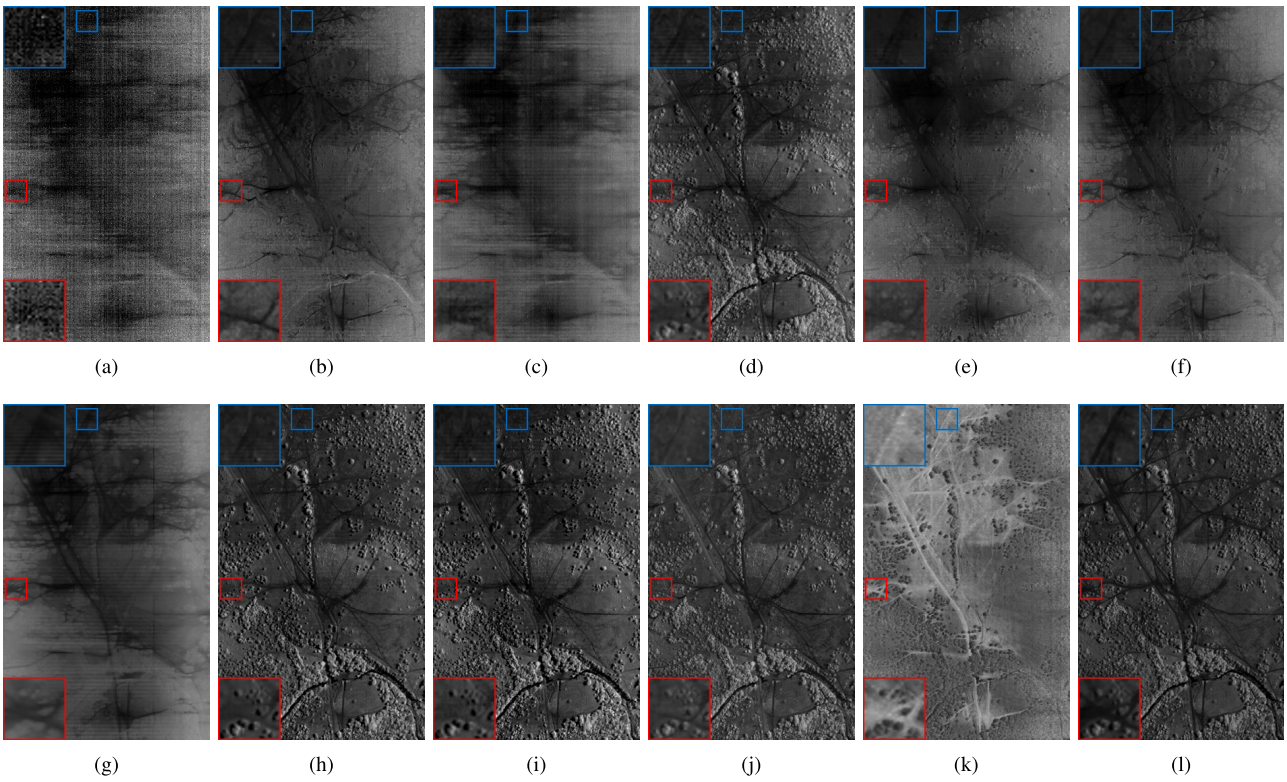


Fig. 8. Denoised HSIs by all the competing methods on band 106 of the Terrain dataset. (a) Observation. (b) NAIRLMA. (c) HyRes. (d) FastHyDe. (e) LRMR. (f) NonLRMA. (g) NGMeet. (h) LRTV. (i) LRTDGS. (j) NMoG. (k) HSIDCNN. (l) BALMF.

noisy image is polluted by strong noise. NAIRLMA, HyRes, FastHyDe, HSIDCNN, and NonLRMA cannot finely recover the truth spectrum. LRMR, LRTV, LRTDGS, and NMoG are able to reconstruct general trend but still suffer from violent fluctuations. As comparison, our proposed BALMF can achieve the closest restoration to the truth spectrum.

To obtain more insights, we show noise histograms for the PCC dataset. Since this is a synthetic noisy image, the real noise is accessible as $\boldsymbol{N} = \boldsymbol{Y} - \boldsymbol{X}$, where $\boldsymbol{Y}$ and $\boldsymbol{X}$ are the noisy image and the clean image, respectively. Given the denoised image $\hat{\boldsymbol{X}}$, we call $\hat{\boldsymbol{N}} = \boldsymbol{Y} - \hat{\boldsymbol{X}}$ the estimated noise. As shown in Fig. 6, real and estimated noise histograms are marked by blue and orange, respectively. Obviously, two histograms are highly coincident, implying that the proposed BALMF could accurately exclude noise from the noisy HSI. Furthermore, the estimated probability density function of the AL distribution is visualized by a blue line, whose asymmetry parameter $\kappa = 0.5713 > 0.5$ corresponds to the fact that real

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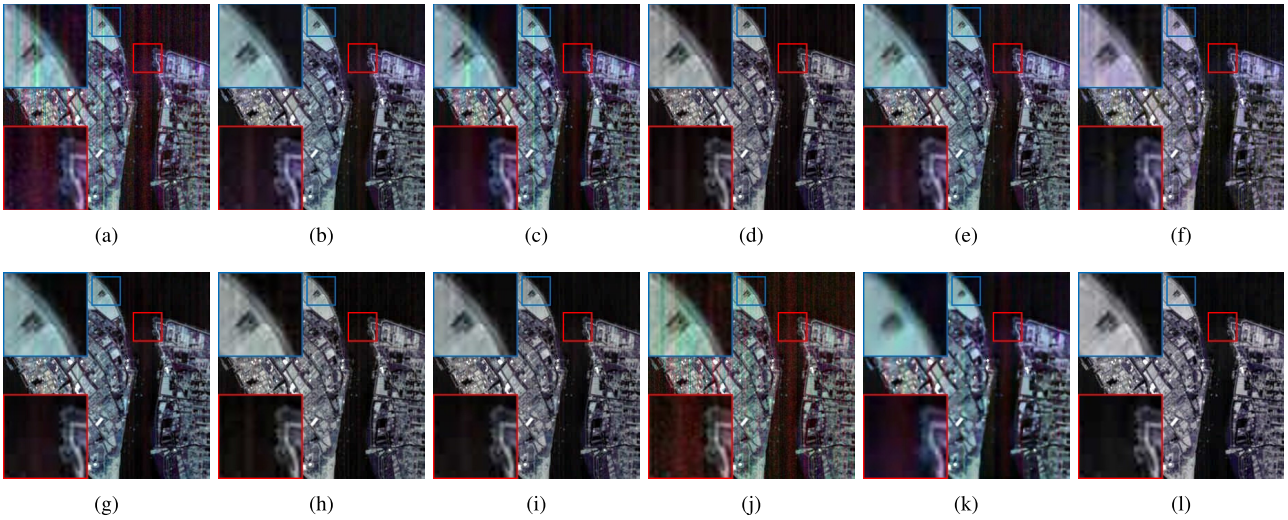


Fig. 9. Denoised HSIs by all the competing methods on band (152, 96, 43) of the Shanghai dataset. (a) Observation. (b) NAIRLMA. (c) HyRes. (d) FastHyDe. (e) LRMR. (f) NonLRMA. (g) NGMeet. (h) LRTV. (i) LRTDGS. (j) NMoG. (k) HSIDCNN. (l) BALMF.

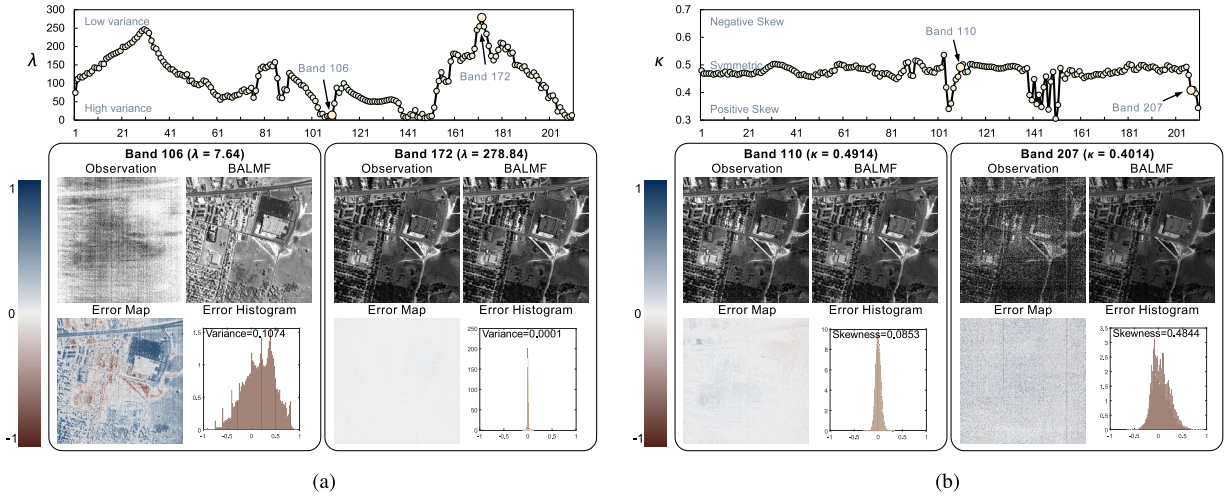


Fig. 10. Analysis on the estimated parameters of the AL distribution. (Top) Estimated parameter curve versus band index. (Bottom) Two representative bands are selected to visualize the noisy band, denoised band by BALMF, error map, and error histogram. (a) Scale parameter λ. (b) Asymmetric parameter κ.

noise is negative skew (i.e., negative values are more frequent than positive values), demonstrating that the AL distribution models synthetic noise well.

B. Real Data Experiments

The HSI denoising algorithms are applied to three real datasets: Urban, Terrain, and Shanghai. The first two datasets were captured by Hyperspectral Digital Imagery Collection Experiment, and the last one was captured by the GaoFen-5 satellite.

The Urban dataset has 307×307 pixels and 210 bands. From Fig. 7, it is observed that band 209 contains severe mixture noise. The image recovered by HSIDCNN has many artifacts. NAIRLMA, HyRes, LRMR, NGMeet, and NonLRMA maintain evident stripes in their recovered images, and FastHyDe, LRTV, and LRTDGS have a small amount of stripes (see the enlarged areas). Comparatively, BALMF and NMoG achieve satisfactory results. The Terrain dataset has 500 $\times$ 307 pixels and 210 bands. In Fig. 8, we can draw the similar conclusion. Yet, NMoG evidently loses many details as shown in the amplified regions. But the image restored by BALMF is still clean and does not contain unpleasant visual effects, manifestly performing more superior than other competing methods. The Shanghai dataset with 300 $\times$ 300 pixels and 155 bands is more challenging, since there are dense stripes and deadlines. To further display details, pseudocolor images composited by bands (152, 96, and 43) are visualized in Fig. 9. It is shown that only BALMF can remove more stripes and deadlines.

C. Analysis on Parameters of AL Distribution

To facilitate understanding on bandwise AL noising modeling, Fig. 10 visualizes estimated scale parameters λ_j and asymmetric parameters κ_j for each band j on the Urban dataset.

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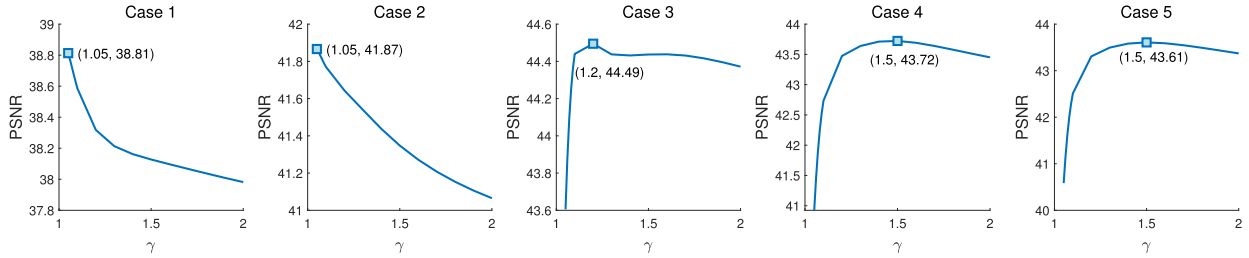


Fig. 11. PSNR curves (versus γ) for Cases 1–5 on the PCC dataset. The point making PSNR maximum is marked by a square, and the coordinate (γ, PSNR) is displayed around this point.

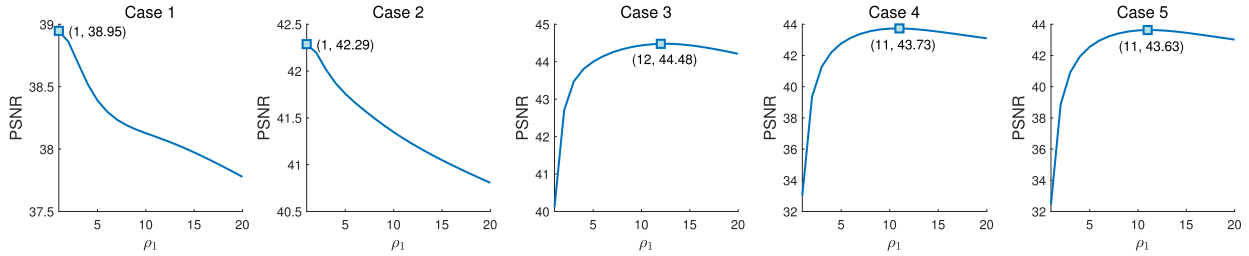


Fig. 12. PSNR curves (versus ρ_1) for Cases 1–5 on the PCC dataset. The point making PSNR maximum is marked by a square, and the coordinate (ρ_1, PSNR) is displayed around this point.

TABLE III
EXECUTION TIME (IN SECONDS) OF ALL COMPETING METHODS

	NAIRLMA	HyRes	FastHyDe	LRMR	NonLRMA	NGMeet	LRTV	LRTDGS	NMoG	BALMF
PCC	13.34	0.74	0.80	12.65	11.14	29.82	7.25	19.10	18.27	17.06
WDCM	37.37	1.39	0.93	15.94	23.59	34.29	15.66	32.86	29.90	33.72
Urban	164.67	4.91	2.42	45.90	329.11	78.70	54.13	114.81	94.28	118.79
Terrain	241.71	9.85	4.02	65.16	593.93	128.64	87.91	182.55	155.56	197.47
Shanghai	83.79	4.81	1.99	26.80	299.83	33.29	69.99	99.43	75.76	58.86

Generally speaking, the higher the scale parameter λ, the smaller the variance. As shown in Fig. 10(a), mild noise is contained in band 172, and BALMF therefore estimates a relatively large-scale parameter $\lambda_{172} = 278.84$; heavy noise corrupts band 106, so the estimated $\lambda_{106} = 7.64$ is comparatively small. Therefore, BALMF can finely estimate the noise density for different bands complying with their real scenarios.

The AL distribution is positive skew if $0 < \kappa < 0.5$ and is negative skew if $0.5 < \kappa < 1$. As shown in Fig. 10(b), band 207 contains Gaussian noise, impulse noise, and stripes, and in the error map positive values are significantly more than negative values. BALMF estimates $\kappa_{207} = 0.4014$, which is consistent with the fact that noise is positive skew. As for band 110, the number of positive and negative values is roughly the same, so $\kappa_{110} = 0.4914$ is very close to 0.5 (that is, approximately symmetric). These results imply that BALMF is good at modeling the asymmetry property of noise.

In summary, bandwise AL noise modeling is able to automatically adapt to noise intensity and is of good effectiveness on general asymmetric and symmetric noise cases.

D. Sensitivity Analysis of Hyperparameters

Many models exploit regularization terms to boost performance, so they have to carefully tune penalty hyperparameters to balance data fidelity and regularization. BALMF focuses on noise modeling and does not use regularization terms, so there is no penalty hyperparameter in our model. From Algorithms 1 and 2, we can find that the main hyperparameters of BALMF are γ, ρ_1, and ρ_2.

1) Parameter γ: As aforementioned, γ is larger than 1 to promote HQS algorithm's convergence. We set $\gamma = 1.05, 1.06, \ldots, 1.09, 1.1, 1.2, \ldots, 2$. Fig. 11 shows how PSNR varies with different γ values for Cases 1–5 on the PCC dataset. It is demonstrated that the PSNR curves constantly decrease as γ grows in Cases 1 and 2, and the PSNR curves increase fast and then decrease slowly as γ grows in Cases 3–5. Accordingly, we can draw the conclusion that BALMF favors small γ in the Gaussian noise scenarios (i.e., Cases 1 and 2), and it favors relatively larger γ in mixture noise scenarios (i.e., Cases 3–5). As a balance between two kinds of scenarios, $\gamma = 1.5$ is recommended.

2) Parameters ρ_1 and ρ_2: Then, let us move on to parameters ρ_1 and ρ_2. In (25), $\boldsymbol{X}$ stays invariant if $\boldsymbol{W}^2$ and ρ_2/ρ_1 scale with equal proportion. Thus, inspired by reference [59], we fix $\rho_2/\rho_1 = \text{mean}(\boldsymbol{W}^2)/2$ and set $\rho_1 = 1, 2, \ldots, 20$. The results are visualized in Fig. 12, and we are able to draw a similar conclusion: BALMF favors small ρ_1 in the Gaussian noise scenarios, but favors relatively larger ρ_1 in the mixture noise scenarios. To strike the balance, $\rho_1 = 10$ is recommended.

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E. Execution Time

Table III lists execution times for all the methods. It is shown that BALMF is slightly slower than LRTDGS, and these methods are ranked from fast to slow as FastHyDe, HyRes, LRMR, LRTV, LRTDGS, BALMF, NAIRLMA, and NonLRMA.

The most time-costing step in BALMF is to solve (18), a weighted LRMF problem with the ℓ_1-norm loss, where we exploit the HQS algorithm and have to implement SVD multiple times. How to further accelerate our method is one of the future work.

V. Conclusion

HSI noise is mostly modeled by symmetric distributions, e.g., Gaussian, Laplacian, and MoG, in the conventional HSI denoising models. But this article discovers that HSI noise is generally asymmetric, which is ignored by previous noise modeling methods. Based on this discovery, we more properly model the HSI noise as bandwise AL distributions and propose a BALMF model. Compared with the representative HSI denoising algorithms, BALMF can better fit HSI noise on both the synthetic and real-world HSIs. Here are some ways to further boost the performance. First, asymmetric noise modeling can be improved by regularizers (e.g., TV). Second, asymmetric noise modeling can be applied to high-order low-rank tensor factorization models avoiding from breaking spatial structure. Finally, we attempt to combine asymmetric noise modeling with deep learning methodologies to further improve the efficiency of the former and the generalization capability of the latter.

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